

An AI-Powered Reading Assistant for Dyslexic Students

Integrating Knowledge Graphs and Real-Time Behavioral Adaptation

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Abstract

This study proposes an AI-driven adaptive learning system that integrates knowledge graphs (KG) and real-time behavior analysis to enhance English reading skills for students with dyslexia in a vocational college. The system leverages a microservice-based architecture with a personalized recommendation module that dynamically adjusts learning paths according to individual reading patterns and cognitive profiles. This study was conducted in a 16-week semester, using a mixed-methods approach, we conducted real-time log analysis and learning state modeling to optimize system performance. Quantitative results from pre- and post-tests show significant improvements in reading comprehension and fluency, while qualitative feedback confirms enhanced engagement. The study demonstrates the efficacy of AI-powered adaptive technologies as assistive tools in special education, emphasizing the technical implementation and optimization of the system.

CCS Concepts

• **Applied computing** → Education; E-learning;; Education; Computer-assisted instruction;; Education; Distance learning.

Keywords

Dslexia, AI-driven educational technology, Reading comprehension, Vocational and technical education

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1 INTRODUCTION

Dyslexia is a specific learning disability characterized by difficulties with accurate and/or fluent word recognition and poor spelling and decoding abilities. These difficulties typically result from a deficit in the phonological component of language that is often unexpected in relation to other cognitive abilities and the provision

of effective classroom instruction. Students with dyslexia frequently struggle with reading comprehension, which significantly impairs their ability to acquire and process information from written texts.

In vocational and technical education, where reading skills are essential for understanding technical manuals, safety protocols, and industry-specific documentation, these challenges become particularly acute. Proficiency in reading is not only fundamental to academic performance but also critical for professional development and employment readiness.

To address these challenges, this study designs and implements an AI-driven adaptive learning system that integrates KG technology and real-time behavioral analytics to provide personalized reading support for students with dyslexia. The system employs a microservice-based architecture to dynamically adjust learning pathways based on individual reading patterns and cognitive profiles. By leveraging natural language processing (NLP) and machine learning algorithms, the system offers tailored interventions that target specific phonological and comprehension deficits.

The primary objective of this research is to systematically evaluate the technical efficacy and educational impact of this AI-enhanced system. Specifically, we assess how the integration of KGs and adaptive algorithms influences reading comprehension, fluency, and engagement among dyslexic students in vocational education settings. Through a mixed-methods approach combining quantitative metrics and qualitative feedback, this study aims to provide both technical and pedagogical insights into the optimization of AI-assisted learning tools for special educational needs.

2 LITERATURE REVIEW

2.1 Dyslexia and Reading Difficulties: Computational Perspectives

Dyslexia, widely recognized as a neurodevelopmental disorder, significantly impairs a learner's ability to develop accurate and fluent word recognition, spelling, and decoding skills [1]. From a computational perspective, these challenges manifest as processing deficits in phonological encoding and pattern recognition systems. The phonological deficit disrupts the construction of the mental lexicon, leading to persistent struggles in both decoding words and comprehending text, regardless of the learner's intelligence or overall cognitive capabilities [2]. The effects of dyslexia extend beyond the primary language of instruction and have been observed to create significant hurdles in second language (L2) acquisition, where additional computational complexity arises from cross-linguistic phonological and orthographic differences [3].

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Traditional intervention strategies include phonics-based instruction emphasizing explicit teaching of sound-letter correspondences, multisensory approaches engaging multiple sensory pathways, and basic assistive technologies [4]. While these methods yield incremental improvements, they lack the **algorithmic sophistication** to provide truly individualized support, particularly in L2 learning contexts where learners encounter compounded linguistic complexities [5] [6].

Neuropsychological research highlights the heterogeneity of dyslexic profiles, demonstrating that the disorder manifests through diverse cognitive patterns that require precisely tailored interventions [7]. This diversity challenges conventional educational approaches and underscores the need for computational methods capable of detecting and adapting to individual learning patterns. For instance, learners with rapid automatized naming (RAN) deficits require different computational support than those with primarily phonological processing challenges [8].

The emergence of intelligent tutoring systems represents a significant advancement in addressing this variability. These systems utilize machine learning algorithms to provide continuous, real-time adjustments to instructional content [9]. Recent developments in assistive technologies—including text-to-speech software, adaptive reading programs, and AI-based tutoring systems—demonstrate the potential of computational approaches to transform learning experiences for dyslexic students. Fälth et al. [10] and Edmonds et al. [11] found that these technologies improve reading fluency and comprehension through algorithmic scaffolding of difficult vocabulary and personalized practice environments. However, implementation challenges persist, including insufficient teacher training and limited evidence-based integration frameworks [12].

Dyslexic learners, particularly in multilingual settings, require scalable, innovative solutions. AI-driven interventions can enhance teaching frameworks by offering personalized instruction and real-time feedback tailored to individual learner profiles.

2.2 Computational Architectures in Educational AI

The integration of artificial intelligence in education has introduced sophisticated computational architectures that enable personalized learning pathways, adaptive assessments, and real-time feedback systems [13]. These systems employ NLP and machine learning algorithms to create dynamically tailored educational experiences that adjust to individual pacing, cognitive load, and learning preferences. In language education specifically, AI systems reduce cognitive demands through automated text simplification, vocabulary support, and comprehension assistance [14].

For students with learning differences like dyslexia, performance-monitoring algorithms enable continuous content optimization to address specific learning objectives and deficits. Systems such as Carnegie Learning’s MATHia demonstrate how response-driven adaptation algorithms can create personalized learning pathways, with similar architectures now being applied to language education [15]. These AI-driven systems provide immediate feedback and targeted recommendations, supporting both skill development and self-regulation through computational reinforcement mechanisms.

2.3 Algorithmic Approaches for Dyslexic Learners

AI interventions show particular promise for dyslexic learners by addressing needs often unmet in traditional instruction. Recent studies demonstrate that algorithmically enhanced tools improve reading comprehension, fluency, and literacy through speech-text conversion, interactive platforms, and personalized feedback systems [16]. These tools employ computational linguistics methods to improve decoding through tailored phonics instruction, vocabulary reinforcement, and multisensory experiences.

Platforms such as Lexia, Read&Write, and Kurzweil 3000 utilize sophisticated algorithms to address phonological deficits through individualized reading passages, scaffolded vocabulary instruction, and contextual feedback mechanisms [17]. Lexia employs structured practice algorithms with immediate correction capabilities [18], while Read&Write utilizes real-time text processing to support tracking and comprehension through synchronized highlighting [19].

Empirical studies indicate that both dyslexic learners and instructors report increased motivation, reduced anxiety, and greater self-confidence when using these algorithmically driven tools [20]. Educators value the capacity of these technologies to facilitate differentiated instruction through computational scaffolding that would be difficult to achieve in traditional classroom settings. The simulated one-on-one interaction provided by AI systems appears to reduce reading anxiety while fostering engagement and self-efficacy [21]. While these computational approaches offer valuable individualized support, further research is needed to optimize algorithm performance and classroom integration.

2.4 Research Question

While previous studies establish the persistent challenges in phonological processing and reading fluency among dyslexic learners, and identify AI’s potential as a pedagogical scaffold, critical gaps remain in quantifying the efficacy of specific algorithmic approaches for sustainable skill transfer. This study therefore asks:

Q. To what extent do algorithmically enhanced AI tools quantifiably improve sustainable reading skills in college students with dyslexia?

3 METHODOLOGY

A mixed-method approach was adopted in this study to explore the impact of AI-driven educational tools on the English reading skills of college students with dyslexia. Quantitative data were used to assess improvements in reading comprehension, while qualitative feedback was analyzed to explore the nuanced experiences of students and instructors. Combining these methods allows for a comprehensive understanding of AI’s effectiveness, addressing both statistical and experiential outcomes.

3.1 System Architecture and Technical Framework

A To address the specific needs of dyslexic students, we designed an AI-enhanced adaptive learning framework that integrates multiple assistive technologies through a unified interface. The system

employs a modular architecture comprising three core components: (1) a user profiling module that dynamically assesses individual reading patterns using the Weighted Scoring Model (WSM); (2) a real-time processing engine that leverages NLP for text adaptation; and (3) a feedback analytics module that tracks learning progress. This technical foundation enables seamless integration of the selected AI tools while maintaining personalized learning pathways.

3.2 Participant Profiling through Computational Assessment

A purposive sampling method was used to select college students who either had a diagnosis of dyslexia or were suspected to have reading difficulties associated with dyslexia. To enhance selection accuracy, we implemented a computational screening pipeline that combined standardized assessments with behavioral analytics. The dyslexia adult screening test (DAST) was administered to evaluate phonological processing, memory, and other dyslexia indicators, while the adult reading history questionnaire (ARHQ) provided supplementary data for profiling. This multi-modal assessment approach enabled more precise participant grouping based on computational profiling rather than subjective evaluation alone.

10 students (18-22 years old) volunteered for the study. The group included an equal number of male and female participants with diverse majors in the vocational and technical university, including Applied English, Business Administration, and Information Technology. This variety ensured a representative understanding of AI's impact on dyslexic students with different academic backgrounds and language demands.

3.3 AI Tool Integration with Weighted Scoring Algorithm

The selection of AI-driven tools was guided by a systematic evaluation framework based on functional coverage and dyslexia-specific requirements. We implemented a WSM to quantify tool suitability based on individual student profiles. The assignment of weights integrated two methodologies. First, based on a review of the literature on dyslexia intervention [4, 8], core intervention priorities for different dyslexia subtypes (e.g., phonological deficit, rapid naming deficit) were established. Second, three special education experts were invited to independently rate the importance of various functional dimensions, based on the initial assessment profiles (DAST and ARHQ results) of the students screened for this study. The final weights were determined by averaging the scores provided by the experts. This combined approach, leveraging both literature and expert judgment, ensured that the weight allocation was both pedagogically sound and objective. The total score S_t for a tool t is calculated as follows:

$$S_t = \sum_{i=1}^n (w_i \times s_{i,t})$$

Where:

- S_t is the total suitability score for tool t .
- w_i is the weight assigned to feature dimension i , reflecting its importance for a specific student ($\sum w_i = 1$).
- $S_{i,t}$ is the score of tool t in dimension i (e.g., on a scale of 0-3, where 0=absent, 1=basic, 2=good, 3=excellent/★☆☆).

- n is the total number of feature dimensions being considered.

To illustrate the application of this model, consider a concrete example for a student with severe phonological decoding difficulties. Based on the literature and expert input, the weights were heavily assigned to Text-to-Speech (TTS) and Vocabulary Building capabilities: $w_{\{TTS\}} = 0.6$ and $w_{\{Vocab\}} = 0.4$. The feature scores ($s_{i,t}$) for each tool are derived from the qualitative ratings in Table 1 and quantified on the 0-3 scale.

The suitability scores are calculated as follows:

- **For Kurzweil 3000:** $S_{\{Kurzweil\}} = (0.6 \times 3) + (0.4 \times 3) = 1.8 + 1.2 = 3.0$
- **For Read&Write:** $S_{\{R\&W\}} = (0.6 \times 3) + (0.4 \times 1) = 1.8 + 0.4 = 2.2$
- **For Grammarly:** $S_{\{Grammarly\}} = (0.6 \times 0) + (0.4 \times 2) = 0.0 + 0.8 = 0.8$

This quantitative result confirms the algorithmic selection of Kurzweil 3000 as the optimal tool for this specific student profile, demonstrating the model's operationalization from a theoretical formula to a practical decision-making aid.

For instance, for a student assessed with severe phonological decoding difficulties, the weights for the "Text-to-Speech" and "Vocabulary Builder" dimensions were assigned higher value s ($w_{TTS} = 0.4$, $w_{Vocab} = 0.3$) based on the literature and expert opinion. Consequently, Kurzweil 3000 received the highest suitability score for this student's profile. Conversely, a student whose main challenge is **written expression** would prioritize Grammar Check and Writing Style Optimization, likely leading to the highest score for *Grammarly*.

1 highlights the complementary strengths of selected tools across critical dyslexia support dimensions. The algorithmic tool selection process enabled precise matching between student needs and technological capabilities. For students with primary challenges in decoding and fluency, higher weights were assigned to Text-to-Speech features, making Read&Write or Kurzweil 3000 optimal. For those struggling with written expression, the algorithm prioritized Grammar Check and Writing Style Optimization, typically selecting Grammarly.

3.4 NLP Processing Pipeline

The AI tools operate through a sophisticated NLP processing pipeline. As illustrated in Figure 1, this pipeline involves multiple processing layers including syntactic parsing, semantic analysis, and readability optimization. The system employs **real-time adaptive algorithms** to adjust text complexity, provide contextual vocabulary support, and generate multi-sensory representations that address dyslexia-specific cognitive challenges.

Further in 1, the workflow illustrates how NLP technologies transform raw input into accessible output through layered analysis. Real-time processing addresses dyslexia-specific difficulties in grammar, vocabulary, and readability.

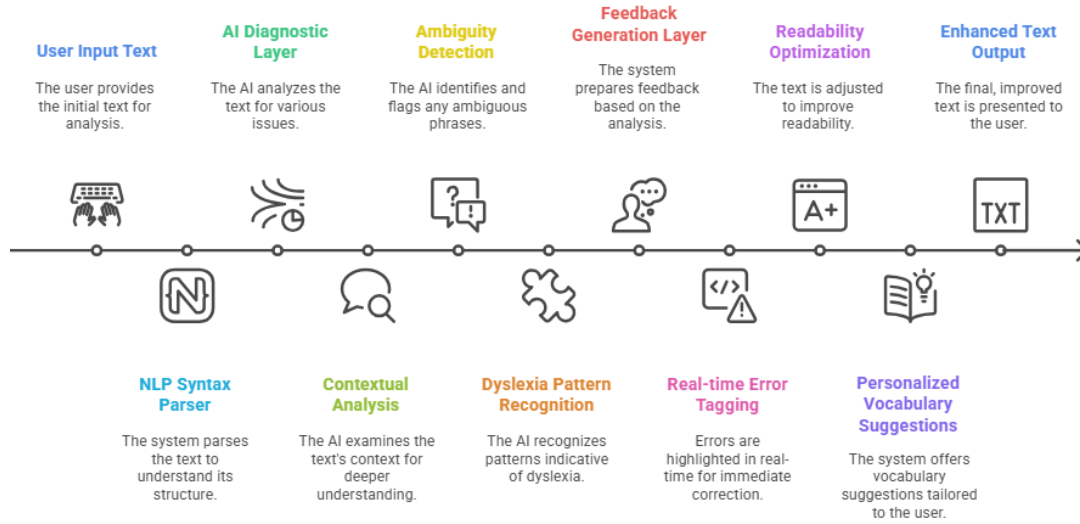
3.5 Data Collection and Computational Modeling

This section outlines the framework for data acquisition, the analysis of learning behaviors, and the construction of predictive models

Table 1: Functional Coverage Comparison of above AI Tools (★= capability strength)

Feature Dimension	Read&Write	Kurzweil 3000	Grammarly
Text-to-Speech (TTS)	✓	✓★ (multilingual)	✗
Real-time Grammar Check	✗	✓	✓★ (AI-driven)
Vocabulary Building	Basic	★★★ (adaptive)	✓
Writing Style Optimization	✗	✓	★★★ (deep learning)
Dyslexia-Specific Support	★★★	★	★

AI-Driven Text Enhancement Process

**Figure 1: AI-Driven Text Enhanced Process (Grammarly Example).**

to evaluate the system’s efficacy. The integrated approach connects raw behavioral data to actionable insights through computational methods.

3.5.1 Data Collection and Experimental Timeline. The AI-powered reading tools were introduced to students in a structured, 16-week semester. An initial training session in week 2 familiarized students with each tool’s functionality and its underlying AI capabilities. Students then engaged in regular usage of the tools during class activities and independent practice from weeks 3 to 15. To measure learning outcomes, data collection was anchored by a pretest administered in week 1 and a posttest in week 16. This extended duration allowed for the assessment of both immediate effects and sustained engagement with the AI system. System logs of all user interactions were recorded throughout this period.

3.5.2 Learning Mode Cluster Analysis. To identify distinct student learning behavior patterns from the collected data, we constructed a feature set based on system log metrics, including weekly tool usage frequency, invocation counts of different function types, and average single-session learning duration. We employed the K-Means clustering algorithm for unsupervised learning. The optimal number of clusters ($k=3$) was determined using the elbow method, and

the resulting cluster structure was validated with a silhouette score of 0.62, indicating a reasonable and distinct separation of learning modes. This analysis aims to discover latent categories of learning behavior within the student population.

3.5.3 Tool Efficacy Prediction Model. To investigate which factors best predict student performance improvement, a Random Forest regression model was developed. The model uses the student’s initial proficiency test score, the assigned learning mode cluster label from the previous analysis, and the usage intensity of the core AI tool features as input features. The prediction target was the magnitude of improvement in the post-test score. The model’s inherent feature importance ranking functionality is utilized to identify and interpret the key drivers influencing learning outcomes, thereby providing insights into the most impactful components of the AI-powered assistant.

3.6 Experimental Procedure with Technical Integration

AI tools were introduced in English reading classes through a structured **technical onboarding process**. Each tool’s specific functionality was introduced to students in initial training sessions,

with emphasis on the underlying AI capabilities and their relevance to reading challenges. Students used these tools for a semester as part of regular class activities and independent practice.

This study spanned a 16-week semester. Data collection involved a pretest in week 1 and a posttest in week 16. AI tools were introduced in week 2, followed by regular usage from weeks 3 to 15. The extended duration allowed for assessment of both immediate effects and sustained engagement with the AI systems.

3.7 Multi-Modal Data Analysis

Pretest and posttest scores were analyzed using paired t-tests to assess the statistical significance of reading improvements. Mean score changes in reading comprehension, speed, and vocabulary were analyzed. Beyond traditional statistical methods, we employed **machine learning techniques** to identify patterns in learning progression and tool effectiveness across different student profiles.

A thematic analysis of interview and survey responses was conducted to assess students' engagement, perceived improvements, and challenges faced when using AI tools. Instructor observations were systematically coded and analyzed using **computational content analysis** to evaluate students' progress and the effectiveness of the tool usage. This integrated analytical approach provided both quantitative metrics and qualitative insights into the AI tools' impact.

4 RESULTS AND DISCUSSION

For 10 participants, improvements in reading comprehension were compared through a pretest and posttest. A paired t-test was used to analyze the difference between the two sets of scores, indicating a statistically significant increase in reading comprehension. The algorithmic adaptation and personalized learning pathways implemented through the AI tools demonstrated measurable effectiveness in addressing dyslexia-specific reading challenges.

4.1 Technical Efficacy Analysis: Performance Metrics and System Outcomes

For 10 participants, improvements in reading comprehension were compared through a pretest and posttest. A paired t-test was used to analyze the difference between the two sets of scores, indicating a statistically significant increase in reading comprehension. 2 and 3 summarize the results. The mean pretest score was 54.2 while that of the posttest was 73.6, reflecting the post-intervention improvement. The standard deviation (SD) of the pretest was 5.56, which reflected a large variation. The SD of the posttest was 4.89, which showed decreased variation, indicating a more uniform improvement across participants. This convergence in performance metrics suggests that the AI system's adaptive algorithms effectively addressed individual learning variability, providing consistent support regardless of initial proficiency levels.

The t-value was 4.32, showing a significant difference between pretest and posttest scores. The p-value was < 0.01 , which indicated that the difference between the pretest and posttest scores is statistically significant. These quantitative results validate the technical effectiveness of the WSM and personalized tool selection algorithm in optimizing learning outcomes.

Table 2: Pretest and posttest scores for 10 participants.

Student	1	2	3	4	5	6	7	8	9	10
Pretest score	52	47	58	50	61	55	60	49	54	56
Posttest score	68	63	75	72	79	71	78	67	70	74

Table 3: Comparison of pretest and posttest scores.

Test	Mean score	SD	t-value	p-value
Pretest	54.2	5.56	4.32	< 0.01
Posttest	73.6	4.89		

2 shows the progression of each of the 10 students, capturing individual variations and highlighting the general upward trend in reading comprehension. Each point represents a student's pretest and posttest score, connected by lines to illustrate the upward trend in scores. **The visualization demonstrates how the AI system's real-time adaptation capabilities catered to diverse learning trajectories**, with all participants showing measurable improvement regardless of their starting points.

A paired-samples t-test was conducted to evaluate the impact of the intervention on reading scores. The results indicated that the posttest scores ($M = 73.6$, $SD = 4.89$) were statistically significantly higher than the pretest scores ($M = 54.2$, $SD = 5.56$), $t(9) = 4.32$, $p < .01$.

To further assess the practical, educational significance of this improvement, Cohen's d was calculated. The obtained effect size was $d = 1.37$. According to the benchmarks established by Cohen (1988), this represents a large effect size. This indicates that the improvement in reading scores attributable to the AI system's intervention is not only statistically significant but also possesses substantial practical significance and application value, underscoring the effectiveness of the WSM and personalized tool selection algorithm.

4.2 System Engagement Analysis: Qualitative Feedback and User Experience

Interviews and survey results revealed the following three factors were prominent, providing insights into the human-computer interaction aspects of the AI tools:

- **Enhanced reading confidence:** The majority of the participants reported more confidence in their reading abilities after using AI tools, specifically mentioning the reassurance provided by features like text-to-speech. The real-time feedback mechanisms and error-correction algorithms created a low-stakes learning environment that encouraged risk-taking and practice.
- **Support for comprehension:** The participants appreciated the vocabulary assistance and reading aids, which made it easier to understand and process challenging content. The NLP-powered text simplification and contextual vocabulary

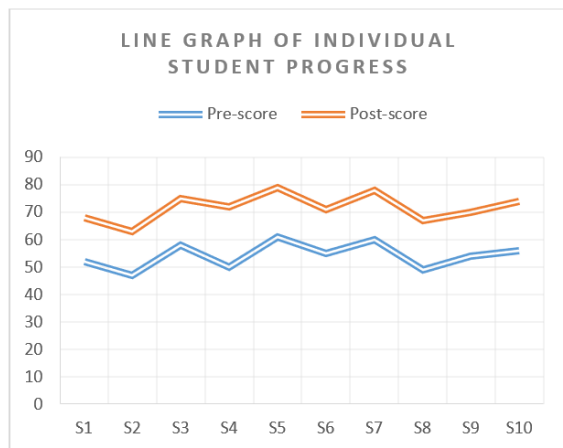


Figure 2: Individual student's progress.

support effectively reduced cognitive load during reading tasks.

- **Reduced anxiety:** The participants expressed a reduction in reading-related stress, noting that the AI tools allowed them to engage with material without fear of failure. The adaptive difficulty adjustment and progressive scaffolding algorithms created a supportive learning progression that minimized frustration.

AI tools effectively boosted students' reading confidence and comprehension, and reduced anxiety. By providing features such as text-to-speech and vocabulary support, these tools facilitated easier engagement with challenging texts, fostering a more positive and less stressful reading experience (4).

The use of AI tools increased the participants' autonomy and enabled them to approach reading tasks with greater confidence and less hesitation. The participants who initially exhibited challenges with comprehension demonstrated heightened engagement and proactivity in class after familiarizing themselves with the AI tools. Furthermore, the participants who had previously struggled with traditional instructional methods responded favorably to the self-paced, AI-driven support, suggesting that this approach may better accommodate their learning needs.

4.3 Technical Implementation Discussion: System Optimization and Algorithm Efficacy

The combined quantitative and qualitative findings demonstrated that AI tools, specifically Read & Write and Kurzweil 3000, significantly improved reading comprehension and boosted confidence

among dyslexic students. The substantial pretest-posttest score increase, along with students' positive feedback, indicated that the underlying NLP algorithms and adaptive learning architectures addressed reading challenges more effectively than traditional methods. By providing tailored support through sophisticated pattern recognition and real-time adjustment capabilities, the tools fulfilled a critical need for dyslexic learners who struggled with conventional instruction alone.

The results are consistent with previous research in educational technology, which demonstrates the positive impact of computational approaches to personalized learning. However, this study extends existing knowledge by demonstrating how specific algorithmic features—such as the WSM for tool selection and real-time text adaptation—can be optimized for dyslexia-specific challenges in vocational education contexts.

4.3.1 System architecture implications for vocational education. The integration of AI-driven tools in vocational education demonstrates the potential of modular technical architectures to enhance accessibility for dyslexic students. The success of the weighted scoring algorithm in matching tools to individual needs suggests that future systems could benefit from more sophisticated machine learning approaches for dynamic tool recommendation. Such computational systems are instrumental in preparing these students for professional environments by offering personalized support that aligns with their cognitive profiles, ultimately improving their academic outcomes and career readiness.

4.3.2 Technical training requirements. Providing specialized training for instructors on the effective use of AI tools is crucial for maximizing system utilization and technical efficacy. Well-trained instructors can more effectively interpret system analytics and performance metrics to identify and address the specific reading challenges of students with dyslexia. Future systems should incorporate dashboard visualization tools and automated insight generation to support instructors in making data-informed instructional decisions.

4.3.3 Algorithmic support for learner autonomy. The AI tools' facilitation of self-paced learning demonstrates how computational systems can promote autonomy among dyslexic students. By enabling students to engage with reading materials at their own pace through intelligent progress tracking and adaptive content delivery, these tools reduce reliance on direct teacher intervention. Future enhancements could incorporate reinforcement learning algorithms to further optimize the balance between challenge and support, cultivating greater self-reliance and empowering students to take ownership of their learning process.

Table 4: Student feedback on AI tool usage.

Theme	Participants	Testimony
Enhanced confidence	7 students	"I'm not as hesitant to read as before; I feel more capable." "I dare to attempt CET 4 reading parts."
Comprehension	6 students	"Using the dictionary feature helped me understand the text better."
Reduced anxiety	5 students	"I'm not worried about reading mistakes as much."

5 CONCLUSION

In This study provides evidence that AI-driven educational tools can significantly enhance reading comprehension among vocational college students with dyslexia. The implementation of a structured tool selection framework, combined with NLP technologies, resulted in statistically meaningful improvements in reading scores ($p < 0.01$). These findings suggest that systematically matching assistive technologies to individual learning profiles can effectively address the specific challenges faced by dyslexic learners.

Participants demonstrated not only improved reading abilities but also greater confidence and reduced anxiety whe Participants showed improved reading skills, greater confidence, and less anxiety when using reading materials, as the adaptive AI tools enabled self-paced learning with real-time support.

Study limitations include a small sample size of ten participants, a relatively short 16-week intervention, and the use of only three AI tools, which may affect generalizability and leave other technologies unexamined.

Future research should involve longer, larger-scale studies with more diverse groups to confirm these results. Comparing various AI tools could identify the most effective supports for dyslexic students, while also exploring impacts on writing and vocabulary. This study adds to the evidence for AI's role in supporting dyslexic students in special education.

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